



cies are working on MEMS sensors for military and defence projects to provide technical solutions for the modern forces. For example, MEMS pressure sensors are used in aircraft, jets, helicopters and various harsh environments; chemical sensors are used to provide accurate and timely information to the soldiers regarding noxious battlefield chemicals; friend-or-foe-identification devices enable soldiers to more easily distinguish their own forces from the enemy's.

**Nano-sensors.** Technical advances in sensor technology from MEMS to nanotechnology have attracted considerable interest for military applications. The main areas of interest for military applications are to improve medical and casualty care; to produce strong, adaptive, lightweight materials for use in clothing, both for protection and to provide enhanced connectivity. Examples are nano-robots and micro-radars for miniature vehicles.

Most of these are under research but are promising prospects not only for civilians but for military and defence applications, too. There are many government and private bodies actively doing research in this field. The list of organisations working on nanotechnology can be found on [https://en.wikipedia.org/wiki/List\\_of\\_nanotechnology\\_organizations#Manufacturers](https://en.wikipedia.org/wiki/List_of_nanotechnology_organizations#Manufacturers)

Nanotechnology materials like carbon nano-tube is a tube-shaped material, made of carbon, having a diameter measuring on nanometre scale. Different nano-particles of varying size, structure and composition are used to make chemi-resistor sensors. These sensors are used for the detection of hazardous gases in military and other security applications.

**Wearable sensors.** Portable and

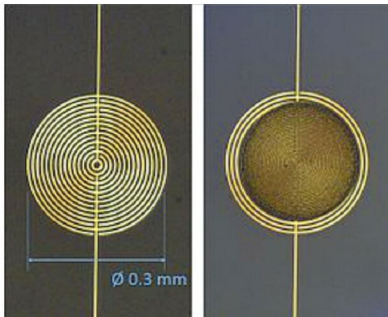


Fig. 5: Circular interdigitated electrodes with (l) and without (r) a gold nanoparticle chemi-resistor film (Credit: <https://en.wikipedia.org>)



Fig. 6: Wearable head-mounted displays (Credit: [www.darpa.mil](http://www.darpa.mil))

wearables devices are most desirable for soldiers. Hence, there has been a growing trend of wearable technologies for wider military applications. These devices are embedded into clothing and accessories that can comfortably be worn on the body. Wearable technology is not about style or fashion but is more about the smart use of technologies to assist soldiers and military operations. These devices should not be confused with headphones, helmets or specially-made shoes/gloves.

Wearable devices with small and smart sensors provide better connectivity, enable easy and real-time transfer of data, images, audios and videos. Wearable technologies can enhance the sensing capability, artificial intelligence (AI)-based solutions, cognitive abilities and augmented reality (AR). Wearable sensors enable soldiers to be tracked easily, especially in high alert operations, to monitor the safety of each soldier and the troop.

### Some leading sensor manufactures for military applications

- Analog Devices
- Cerion
- Eaton Corp.
- General Electric
- Honeywell International
- Kulite Semiconductor Products
- Leonardo DRS
- NXP Semiconductors
- Reynard Corp.
- Safran Group
- STMicroelectronics
- Texas Instruments
- Viooa Imaging Technology
- Xenics
- Zyxex Technologies

### To sum up

The concept of battlefield and military operations has changed significantly over the years due to technological revolutions. The nature of support provided by sensor technologies could be wide ranging from simple sensors to AI-based tools for real-time intelligence sharing and decision support. From soldiers on ground to combat pilots, sensors are making their relevance in various military applications. However, with continuous technological advancements in the fields of wireless sensors, the Internet of Things (IoT), robotics, cognitive sciences, image sensors, wearables and nanotechnology, sensors have a huge potential and can be explored further for military applications. **EFY**